

CITYCARE GENERAL HOSPITAL

SURGICAL SERVICES STANDARD OPERATING PROCEDURE

Pre-operative, Intra-operative & Post-operative Care

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Prepared By	Quality Assurance & Clinical Governance Team
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1. PURPOSE & SCOPE

This SOP establishes the clinical and safety standards for all surgical procedures performed at CityHospital, from pre-operative assessment through to post-operative recovery and ward care. It applies to all elective, urgent, and emergency surgical cases across all surgical specialties.

- Ensure systematic pre-operative patient optimisation and surgical risk stratification.
- Mandate WHO Surgical Safety Checklist compliance for every case without exception.
- Maintain the highest standards of sterile technique and intra-operative patient safety.
- Prevent surgical site infections, anaesthetic complications, and wrong-site surgery.
- Facilitate safe and smooth patient transition from theatre to PACU to ward.
- Comply with national surgical safety standards and accreditation requirements.

2. DEFINITIONS & ABBREVIATIONS

- **PACU:** Post-Anaesthetic Care Unit — the recovery room immediately after surgery.
- **ASA Classification:** American Society of Anaesthesiologists physical status scale (I–VI).
- **WHO Checklist:** World Health Organization Surgical Safety Checklist with three phases: Sign In, Time Out, and Sign Out.
- **SIP:** Surgical Infection Prophylaxis — prophylactic antibiotic administered before skin incision.
- **DVT:** Deep Vein Thrombosis — surgical patients at elevated risk; thromboprophylaxis mandatory.
- **TIVA:** Total Intravenous Anaesthesia.
- **RSI:** Rapid Sequence Induction — emergency airway management technique.
- **NMB:** Neuromuscular Blockade — muscle paralysis to facilitate intubation and surgical conditions.
- **SSI:** Surgical Site Infection — a preventable complication closely linked to perioperative care quality.
- **Aldrete Score:** Post-anaesthetic recovery scoring system used for PACU discharge decision.

3. ROLES & RESPONSIBILITIES

Role	Department	Responsibilities
Surgeon / Operator	Surgery	Surgical procedure, consent, post-op orders, SSI prevention, discharge planning
Anaesthetist	Anaesthesia	Pre-op assessment, anaesthesia delivery, airway management, PACU handover
Scrub Nurse / ODP	Theatre Nursing	Sterile field, instrument counts, WHO checklist, specimen management
Circulating Nurse	Theatre Nursing	Equipment, specimen labelling, documentation, communication
PACU Nurse	PACU	Post-op monitoring, pain management, discharge assessment (Aldrete)
Theatre Coordinator	Theatre Management	Surgical list management, equipment booking, team safety briefing
Pre-op Assessment Nurse	Pre-op Clinic	Nursing pre-assessment, consent verification, preparation instructions

4. PRE-OPERATIVE ASSESSMENT & PREPARATION

4.1 Pre-Operative Assessment Clinic (at least 48 hours before surgery)

- Full medical history, current medications, allergies, and previous anaesthetic history reviewed.
- ASA physical status classification assigned; high-risk patients referred to specialist optimisation clinics.
- Investigations ordered based on ASA class and surgery type: FBC, U&E, coagulation, ECG, CXR as clinically indicated.
- Informed surgical and anaesthetic consent obtained and witnessed; signed copy filed in EMR.
- Anticoagulant and antiplatelet medication bridging plan documented and communicated to patient.
- DVT risk stratified using Caprini score; prophylaxis type selected and documented.
- Blood group and screen sent; cross-match ordered for major vascular, cardiac, or complex orthopaedic cases.
- Nil-by-mouth instructions provided in writing: 6 hours for solids, 2 hours for clear fluids.

4.2 Day of Surgery Preparation Checklist

- Step 1.** Patient identity confirmed by nurse using 3-point check: full name, date of birth, and MRN. Identity band applied and verified.
- Step 2.** Surgical site marked by the operating surgeon in indelible pen with the patient awake, alert, and confirming.
- Step 3.** Pre-operative checklist completed: consent present and signed, site marked, allergies confirmed, implant availability verified.
- Step 4.** Surgical infection prophylaxis antibiotic administered 30 to 60 minutes before skin incision; documented in EMR.
- Step 5.** DVT prophylaxis applied: graduated compression stockings and LMWH as per pre-assessment plan.
- Step 6.** Patient transferred to anaesthetic room; full haemodynamic monitoring applied; IV access confirmed and patent.

5. WHO SURGICAL SAFETY CHECKLIST

WHO Surgical Safety Checklist compliance is MANDATORY for every surgical procedure at CityHospital without exception. Non-completion must be reported as a clinical incident.

5.1 SIGN IN — Before Anaesthesia Induction

- Patient verbally confirms identity, site of procedure, and that consent has been signed.
- Anaesthesia safety check fully completed; pulse oximeter functioning and displayed.
- Known allergies confirmed by patient and verified against EMR.
- Difficult airway or aspiration risk identified; management plan in place and communicated.

5.2 TIME OUT — Before Skin Incision (entire team STOPS)

- Surgeon, anaesthetist, and scrub nurse each verbally confirm patient identity, procedure name, and operative site.
- Antibiotic prophylaxis confirmed administered within the past 60 minutes.
- Surgeon communicates critical or non-routine steps, anticipated blood loss, and any special equipment needs.
- Anaesthetist confirms patient-specific concerns.
- Nursing team confirms equipment sterility, implant availability, and any equipment concerns.
- Imaging displayed and confirmed correct patient if relevant to the procedure.

5.3 SIGN OUT — Before Patient Leaves Theatre

- Procedure name recorded accurately and confirmed verbally.
- Swab, needle, and instrument counts confirmed correct by both scrub and circulating nurse.
- Specimens labelled accurately in the patient's presence; pathology request form completed.
- Any equipment problems requiring report documented.
- Key recovery and post-operative management concerns communicated to PACU nurse.

6. INTRA-OPERATIVE STANDARDS

- Strict sterile field maintained throughout the procedure; any breach immediately reported to scrub nurse and surgeon.
- All sharps handled using the neutral zone technique; hand-to-hand sharps passing is strictly prohibited.
- Blood loss monitored continuously; cell salvage activated for major vascular and complex orthopaedic cases.
- Intra-operative core body temperature maintained at or above 36 degrees C using forced-air warming blanket and warmed IV fluids.
- Patient positioning: all pressure points padded; nerve stretch prevented; position documented in the anaesthetic record.
- Surgical specimens: double-labelled with patient MRN and site; appropriate fixative container; pathology form completed.
- Instrument, swab, and needle count performed: at case opening, before closing any body cavity, and at wound closure.

WARNING: NEVER close wound until ALL counts are confirmed CORRECT by both scrub and circulating nurse independently. Any discrepancy mandates immediate surgical exploration and radiology before closure.

7. ANAESTHESIA PROTOCOL

7.1 Induction

- Pre-oxygenation for minimum 3 minutes before induction; ETCO₂ monitoring applied before intubation.
- Induction agent selected based on ASA class, cardiac status, gastric risk, and airway assessment.
- Difficult airway plan communicated before induction; video laryngoscope first-line for predicted difficult intubation.
- Correct ETT placement confirmed by continuous ETCO₂ waveform plus bilateral auscultation; recorded in EMR.
- Skin incision not made until anaesthetist confirms patient is adequately anaesthetised and analgesed.

7.2 Maintenance

- Depth of anaesthesia monitoring (BIS or entropy) for all TIVA cases and high-risk patients.
- Muscle relaxation monitored by train-of-four (TOF) stimulation throughout case.
- Full reversal of neuromuscular block confirmed (TOF ratio above 0.9) before extubation.
- PONV prophylaxis: ondansetron and dexamethasone for high-risk patients (Apfel score 2 or above).
- Fluid management: balanced crystalloid; vasopressor use for refractory hypotension; goal-directed fluid therapy for major cases.

7.3 Emergence & Extubation

- Awake extubation preferred: patient responsive to commands, coughing, and protective reflexes intact.
- Head-down or lateral position for high aspiration risk patients during emergence.
- Immediate post-extubation: supplemental oxygen, continuous SpO₂ monitoring, verbal responsiveness confirmed.

8. POST-ANAESTHETIC CARE UNIT (PACU)

- Step 1.** Anaesthetist provides structured SBAR verbal handover to PACU nurse before leaving the department.
- Step 2.** Continuous monitoring applied: SpO2, ECG, non-invasive BP every 5 minutes initially, then every 15 minutes when stable.
- Step 3.** Pain assessed using NRS at arrival; multimodal analgesia titrated to achieve NRS score of 3 or below.
- Step 4.** PONV assessed; antiemetics administered as per PACU protocol; patient positioned appropriately.
- Step 5.** Aldrete score assessed every 15 minutes; discharge to ward when score of 9 or above on two consecutive assessments.
- Step 6.** Surgeon or anaesthetist notified if Aldrete score not achieved within 90 minutes of PACU arrival.
- Step 7.** Written PACU nursing summary completed; verbal SBAR handover to ward nurse at time of transfer.

Aldrete Parameter	Score 2	Score 1	Score 0
Activity	Moves 4 limbs	Moves 2 limbs	Moves 0 limbs
Respiration	Deep, coughs	Dyspnoeic but adequate	Apnoeic
Circulation	BP +/-20% baseline	BP +/-20–50% baseline	BP +/-50% baseline
Consciousness	Fully awake	Arousable	Not responding
SpO2	Above 92% on air	Needs O2 to stay above 90%	Below 90% with O2

9. POST-OPERATIVE WARD CARE

- Vital signs monitoring: every 15 minutes for 1 hour, every 30 minutes for 2 hours, hourly for 4 hours, then 4-hourly if stable.
- Wound assessment: inspect dressing at 1 hour and 4 hours post-operatively; notify surgeon if soakage exceeds 50% of dressing.
- Fluid balance: strict input and output charting; oral fluids reintroduced per surgeon's orders.
- Mobilisation: physiotherapist review within 24 hours; early mobilisation reduces DVT, pulmonary embolism, and ileus.
- DVT prophylaxis: LMWH first dose within 6 to 12 hours of surgery; continue until patient fully mobile.
- Drain management: document drainage volume hourly for first 4 hours; notify surgeon if exceeds 100 ml/hr.
- Post-operative analgesia: multimodal approach (regular paracetamol, NSAID if no contraindication, opioid as rescue only); PCA for major cases.
- Nausea management: regular antiemetic for first 24 hours in high-risk patients; oral intake as tolerated.
- Blood glucose monitoring: 4-hourly for all diabetic patients and patients on steroid therapy.

10. SSI PREVENTION, EXCEPTIONS & BEST PRACTICES

10.1 Surgical Site Infection Prevention Bundle

- Pre-operative MRSA screening; nasal decolonisation with mupirocin for positive patients (5 days pre-operatively).
- Hair removal by clipping only within 2 hours of surgery — razor shaving is prohibited due to increased SSI risk.
- Skin antisepsis with chlorhexidine-alcohol; full dry time observed before surgical draping.
- Normothermia maintained throughout the perioperative period; hypothermia is an independent SSI risk factor.
- Perioperative blood glucose maintained below 10 mmol/L in all diabetic patients.
- Wound closure: glove change after contaminated phase of surgery before closing wound.

10.2 Exceptions & Special Circumstances

- Emergency surgery: pre-operative assessment compressed to essential elements; consent by phone if patient unconscious (two-physician rule applies).
- Paediatric surgery: weight-based drug dosing mandatory using Broselow or verified body weight; paediatric anaesthetist required for all patients under 2 years.
- Day-case surgery discharge: Aldrete score 9 or above, tolerating oral fluids, responsible adult escort confirmed, written emergency contact provided.

10.3 Best Practices

- Pre-operative team briefing: surgeon leads 5-minute team briefing before first case of the day; unusual steps, risk alerts, and equipment needs discussed.
- Lean theatre workflow: all instruments, consumables, and imaging verified before patient arrives in theatre.
- Enhanced Recovery After Surgery (ERAS) protocol implemented for all elective colorectal, orthopaedic, and major abdominal cases.